

Check-In 08/20. (True/False) The integral $\int (x-1)^2(x+5)^3 dx$ can be computed as a ‘shifting’ integral.

Solution. The statement is *false*. One wants to multiply the two terms in the integrand without explicitly expanding—but the non-matching powers prevents this. However, this might be possible with a ‘shift.’ There are only two possible choices for u to ‘shift’ the integrand: $u = x - 1$ (so that $x = u + 1$) or $u = x + 5$ (so that $x = u - 5$). In both cases, we have $du = dx$. In the first case, the resulting integral is $\int u^2(u+6)^3 du$, and in the second case, the resulting integral is $\int (u-6)^2 u^3 du$. In either case, we still cannot simply multiply the integrands without explicit expanding. To compute the given integral, there are several options. The most obvious is simple expansion:

$$\begin{aligned}\int (x-1)^2(x+5)^3 dx &= \int (x^5 + 13x^4 + 46x^3 - 10x^2 - 175x + 125) dx \\ &= \frac{x^6}{6} + \frac{13x^5}{5} + \frac{23x^4}{2} - \frac{10x^3}{3} - \frac{175x^2}{2} + 125x + C\end{aligned}$$

Alternatively, we can use integration-by-parts—particularly tabular integration (as each possible term in the integrand differentiates to 0 and can be repeatedly integration). The ‘harder’ choice of tabular integration yields...

u	dv
$(x+5)^3$	$(x-1)^2$
$3(x+5)^2$	$\frac{(x-1)^3}{3}$
$6(x+5)$	$\frac{(x-1)^4}{12}$
6	$\frac{(x-1)^5}{60}$
0	$\frac{(x-1)^6}{360}$

Therefore, we have...

$$\int (x-1)^2(x+5)^3 dx = \frac{(x+5)^3(x-1)^3}{3} - \frac{(x+5)^2(x-1)^4}{4} + \frac{(x+5)(x-1)^5}{10} - \frac{(x-1)^6}{60} + C$$

Alternatively, we have...

u	dv
$(x-1)^2$	$(x+5)^3$
$2(x-1)$	$\frac{(x+5)^4}{4}$
2	$\frac{(x+5)^5}{20}$
0	$\frac{(x+5)^6}{120}$

Therefore, we have...

$$\int (x-1)^2(x+5)^3 dx = \frac{(x-1)^2(x+5)^4}{4} - \frac{(x-1)(x+5)^5}{10} + \frac{(x+5)^6}{60} + C$$

Observe that none of these answers are equal to each other but instead differ by a constant.

Check-In 08/25. (True/False) To integrate $\int xe^{-3x^2} dx$, one uses LIATE to choose $u = x$ and $dv = e^{-3x^2}$.

Solution. The statement is *false*. To compute this integral with this choice of u and dv , one would have to integrate e^{-3x^2} . However, the function e^{-x^2} (and hence e^{-3x^2} , which differs only by a linear u -substitution of $x = \sqrt{3}u$, i.e. $u = x/\sqrt{3}$) has no elementary antiderivative, i.e. is not integrable. So, this approach will not work. However, we can integrate this with a simple u -substitution. Choosing $u = -3x^2$, we have $du = -6x dx$. But then...

$$\int xe^{-3x^2} dx = -\frac{1}{6} \int -6xe^{-3x^2} dx = -\frac{1}{6} \int e^u du = -\frac{e^u}{6} + C = -\frac{e^{-3x^2}}{6} + C$$

We can easily verify this as $\frac{d}{dx} \left(-\frac{e^{-3x^2}}{6} + C \right) = -\frac{e^{-3x^2}}{6} \cdot -6x + 0 = xe^{-3x^2}$.

Check-In 08/27. (True/False) The integral $\int \sin(2x) \cos(3x) dx$ can be treated as a 'looping integral.'

Solution. The statement is *true*. We can choose either $u = \sin(2x)$ or $u = \cos(3x)$. Choosing $u = \sin(2x)$, we have $dv = \cos(3x)$. We then have...

u	dv
$\sin(2x)$	$\cos(3x)$
$2 \cos(2x)$	$\frac{\sin(3x)}{3}$
$-4 \sin(2x)$	$-\frac{\cos(3x)}{9}$

So that...

$$\int \sin(2x) \cos(3x) dx = \frac{\sin(2x) \sin(3x)}{3} + \frac{2 \cos(2x) \cos(3x)}{9} + \frac{4}{9} \int \sin(2x) \cos(3x) dx$$

$$\frac{5}{9} \int \sin(2x) \cos(3x) dx = \frac{\sin(2x) \sin(3x)}{3} + \frac{2 \cos(2x) \cos(3x)}{9}$$

$$\int \sin(2x) \cos(3x) dx = \frac{9}{5} \left(\frac{\sin(2x) \sin(3x)}{3} + \frac{2 \cos(2x) \cos(3x)}{9} \right) + C$$

$$\int \sin(2x) \cos(3x) dx = \frac{1}{5} (3 \sin(2x) \sin(3x) + 2 \cos(2x) \cos(3x)) + C$$

Alternatively, choosing $u = \cos(3x)$ and $dv = \sin(2x)$, we have...

u	dv
$\cos(3x)$	$\sin(2x)$
$-3 \sin(3x)$	$-\frac{\cos(2x)}{2}$
$-9 \cos(3x)$	$-\frac{\sin(2x)}{4}$

So that...

$$\int \sin(2x) \cos(3x) dx = -\frac{\cos(3x) \cos(2x)}{2} - \frac{3 \sin(3x) \sin(2x)}{4} + \frac{9}{4} \int \cos(3x) \sin(2x) dx$$

$$-\frac{5}{4} \int \sin(2x) \cos(3x) dx = -\frac{\cos(3x) \cos(2x)}{2} - \frac{3 \sin(3x) \sin(2x)}{4}$$

$$\int \sin(2x) \cos(3x) dx = -\frac{4}{5} \left(-\frac{\cos(3x) \cos(2x)}{2} - \frac{3 \sin(3x) \sin(2x)}{4} \right) + C$$

$$\int \sin(2x) \cos(3x) dx = \frac{1}{5} (2 \cos(3x) \cos(2x) + 3 \sin(3x) \sin(2x)) + C$$

Alternatively, one can use the product-to-sum formula:

$$\sin A \cos B = \frac{\sin(A+B) + \sin(A-B)}{2}$$

We then have...

$$\begin{aligned} \int \sin(2x) \cos(3x) dx &= \int \frac{\sin(2x+3x) + \sin(2x-3x)}{2} dx \\ &= \frac{1}{2} \int (\sin(5x) + \sin(-x)) dx \\ &= \frac{1}{2} \left(-\frac{\cos(5x)}{5} + \cos(-x) \right) + C \\ &= \frac{1}{2} \left(\cos x - \frac{\cos(5x)}{5} \right) + C \\ &= \frac{1}{10} (5 \cos x - \cos(5x)) + C \end{aligned}$$